

Invoice Portal Admin

Architecture, user experience & AI integration

A 30-minute executive briefing
Primary presenter + database colleague

Operate

Find and maintain invoices in one workspace.

Ask

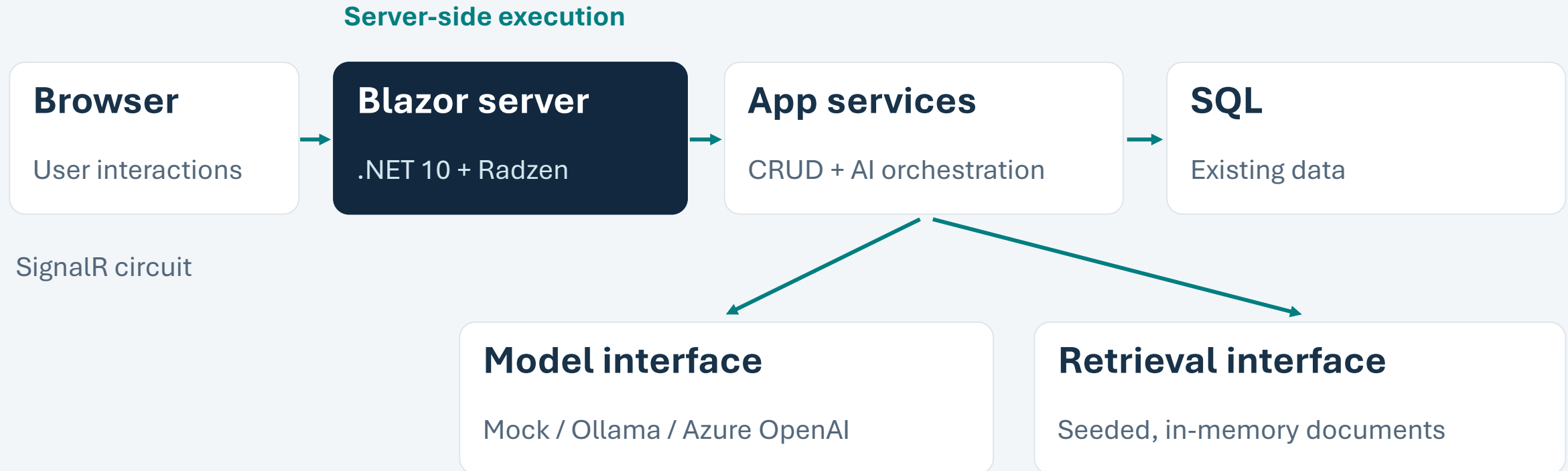
Explore reporting and document Q&A through shared AI interfaces.

Verify

Inspect results, sources, and the controls still needed for production.

Understand the operating experience, its data foundation, and the boundary between demonstration and production.

How the system fits together



The server connects the user experience to SQL and two replaceable AI interfaces.

One workspace for invoice administration

ILLUSTRATIVE • SYNTHETIC CONTENT • NOT A SCREENSHOT

INVOICE PORTAL ADMIN

Invoices

FILTER / SORT / COLUMNS

Invoice	Currency	Status
[Synthetic invoice]	[Code]	[State]
[Synthetic invoice]	[Code]	[State]

Detail tabs: Header | Parties & program | Submitter | Revision & SAP

01 Find / filter

Synthetic invoice • Currency: [code] • Status: [state] • Filter / Sort

02 Inspect / edit

Header | Parties & program | Submitter | Revision & SAP

03 Assistance

AI Insights / Ask documents • Inspect before acting

ILLUSTRATIVE editable wireframe with synthetic content — not a screenshot or evidence of live records.

AI Insights: a question becomes a report

1 • Ask

Question + schema description → model prompt



2 • Propose

Model → structured SQL and parameter values

3 • Check / execute

JSON validation → SQL guard → bounded database execution



4 • Inspect

Result rows → UI grid; generated SQL can be expanded

The model proposes SQL; application code validates and executes it, then presents result rows to the user.

Document answers with visible sources

Retrieve

Seeded fictional documents → in-memory BM25 + customer relevance

Answer

Question + selected passages → model answer

Verify

Retain recognized citations; inspect excerpts against the answer

Provider seam

Mock / Ollama / AzureOpenAI exist; active runtime provider is unverified

Seeded documents demonstrate retrieval and citations; neither source labels nor customer relevance guarantee correctness or access.

Live walkthrough: find, ask, verify

Orient · 0:00–0:45

Dashboard and visible provider;
approved data only

Find · 0:45–2:00

Filter / sort; inspect edit dialog; close
without saving

Ask · 2:00–3:30

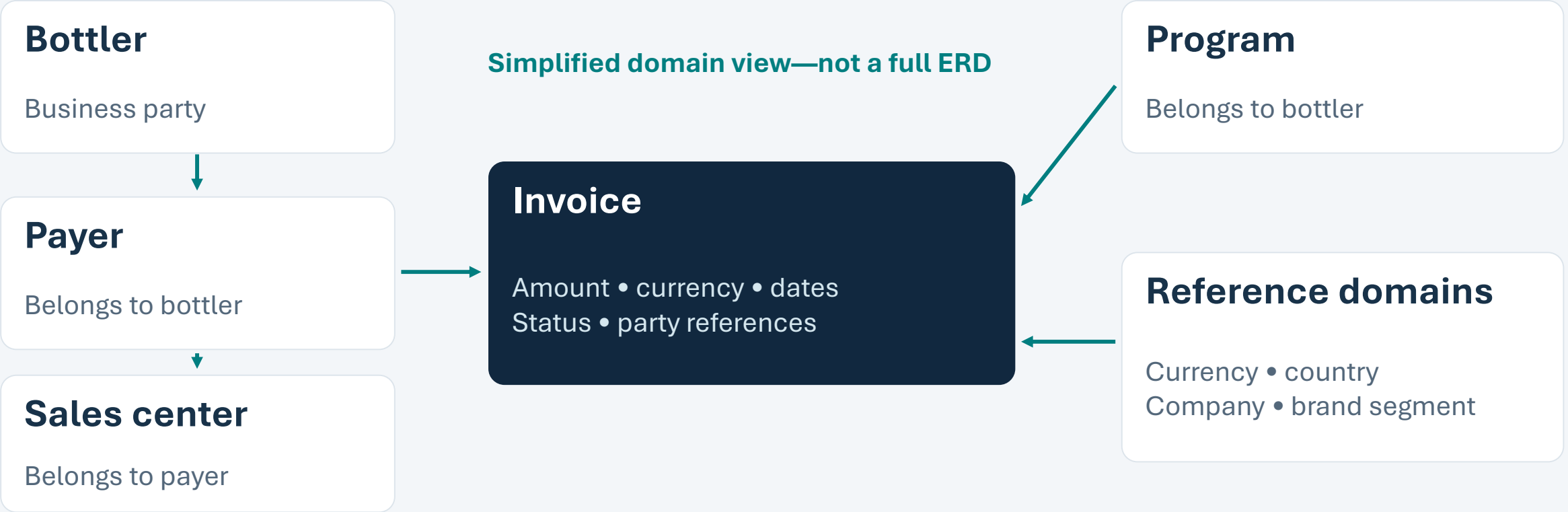
Count invoices per currency; inspect
results and generated SQL

Verify · 3:30–5:00

Seeded document question, citation,
excerpt; hand off to Database

Show the approved experience without saving changes; disclose the provider and verify every displayed answer.

The business data behind the portal



Invoices connect business parties and reference domains; accurate reporting depends on those relationships and definitions.

How database functionality reaches users

Grid request

Radzen filters, sort and page → shared CRUD service



Database work

Context per operation → SQL count + bounded page → grid

Related choices

Lookup projections + short circuit-local cache → dropdowns



Save / feedback

Detached edits → SQL constraints → translated database errors

Shared CRUD and lookup services connect server-side invoice grids and relationship-aware dialogs to SQL.

Integrity, lifecycle, and environments

01

Integrity

Relationships,
uniqueness, timestamps
and computed status

02

Lifecycle

Soft delete / restore
where supported; not a
complete audit trail

03

Separate environments

Local bacpac import ≠
existing Azure SQL;
metadata sanitation ≠
anonymization

Existing constraints and lifecycle behavior support administration; local import does not anonymize business data.

Controls for AI-assisted reporting

01

Built • validation

Structured response checks, SELECT/WITH intent, denylist and parameter checks

02

Built • bounded execution

Circuit rate limit, command / lock timeouts, returned-row cap and rollback

03

Required • SQL boundary

Separate AI read identity + approved objects + enforced user data scope

Application checks are defense in depth; a separate least-privilege AI database reader is not implemented.

Hosting and operational visibility

01

Local view

App + SQL Server; DbInit;
optional Aspire telemetry
/ Ollama

02

Azure template · unverified

Container Apps → existing
SQL; managed identity,
optional OpenAI, Blob key
ring

03

Observe / qualify

OpenTelemetry traces /
metrics; Serilog logs;
conditional sinks and
Easy Auth

Local tooling and Azure templates provide an operating foundation, not evidence of a deployed or resilient production service.

What the demonstration establishes

01

Administration

Invoice and reference-data workflows are implemented.

02

AI integration

Reporting and document Q&A share replaceable interfaces.

03

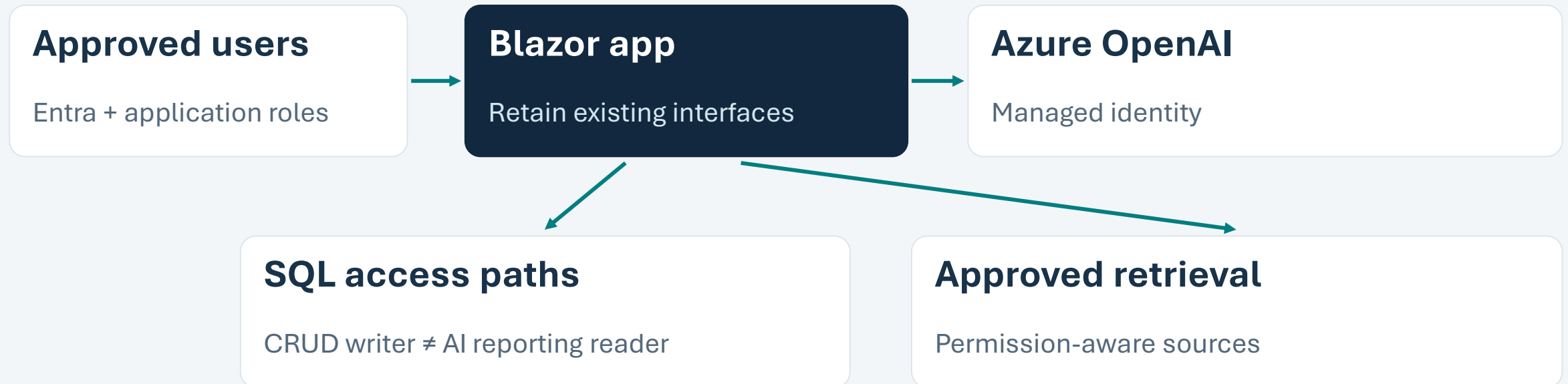
Next distinction

Capability evidence is not production acceptance.

The integration exists; what separates this demonstration from production?

Azure OpenAI: from integration to production

PROPOSED TARGET • NOT DEPLOYED



Built: client + identity wiring | **Required:** access policy | **Validate:** model, region, capacity

TARGET, not deployed: retain the client interfaces while enforcing user access and approved model, data and network boundaries.

Two AI workflows. Two data boundaries.

01

Reporting · DBA + App

Separate read identity;
curated views; enforced
user / customer / column
scope

02

Documents · Data owner + App

Approved sources; ACL-
aware ingestion /
retrieval; freshness,
deletion and provenance

03

Both · Security + AI

Minimize data • Review
logging • Evaluate quality
/ leakage • Human review

Reporting needs authorized database access; document Q&A needs authorized, current sources; both need privacy and output review.

Go live through evidence—not configuration

1 · Secure foundation

Security + Platform +
DBA: identity, data scope,
model and network
approval



2 · Limited pilot

App / AI + data owners:
correctness,
authorization, leakage
and human review



3 · Operated rollout

Operations + Platform +
Finance: recovery,
capacity, monitoring and
spend controls

Governance decisions unblock implementation; evaluated data paths and operational signoff determine rollout readiness.

Discussion

01

Operational fit

Which invoice tasks and questions would a limited pilot need to support?

02

Governance

Who may access which records and sources, and who reviews AI outputs?

03

Readiness priorities

Which security, quality and recovery evidence must precede rollout?

Agree operational fit, governance expectations and the next evidence to collect—without implying approval to deploy.

Invoice workspace

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Detail tabs: Header | Parties & program | Submitter | Revision & SAP

01 Grid • synthetic content

[Synthetic invoice] | [currency] | [status] •
Filter / Sort / Columns

02 Dialog • inspect only

Header | Parties & program | Submitter |
Revision & SAP • Close without saving

03 Navigation

Dashboard → Invoices → AI Insights / Ask
documents

ILLUSTRATIVE editable wireframe with synthetic content — not a screenshot, live record or completed demonstration.

AI reporting

ILLUSTRATIVE • SYNTHETIC CONTENT • NOT A SCREENSHOT

INVOICE PORTAL ADMIN

AI Insights

Count invoices per currency

Generated SQL ▶ [\[expand to inspect\]](#)

Currency	InvoiceCount	TotalAmount
[Code]	[Count]	[Synthetic amount]

01 Question

Count invoices per currency • Provider: [verify at rehearsal]

02 Synthetic result grid

Currency: [code] | InvoiceCount: [count] | TotalAmount: [synthetic amount]

03 Inspect

Generated SQL [collapsed / expand] • Result filter • Returned-row limit

ILLUSTRATIVE editable report with synthetic placeholders — no query was run and no result values are asserted.

Answers with visible sources

ILLUSTRATIVE • SYNTHETIC CONTENT • NOT A SCREENSHOT

INVOICE PORTAL ADMIN

Ask the documents

What supplies are listed in this illustrative source?

The illustrative source lists sample supplies. [S1]

S1 / SYNTHETIC SOURCE
Excerpt: Sample supplies are listed here.

01 Synthetic question

What supplies are listed in this illustrative source?

02 Synthetic answer

The illustrative source lists sample supplies. [S1]

03 Synthetic source • S1

Illustrative excerpt: Sample supplies are listed here. • Show / Hide excerpt

ILLUSTRATIVE editable wireframe with synthetic content — citation labels support inspection, not proof of correctness.

Technical sources and production limitations

01

Application + data

UI, service and schema sources; lifecycle and integration limits

02

AI + privacy

Prompts, query checks, citations and log privacy

03

Platform + assurance

Infrastructure, telemetry, CI and official production guidance

Source evidence describes implemented behavior; production guidance and acceptance work do not establish deployment readiness.